

MATH 345 Skeleton Notes

Unit 1: Vectors, matrices, and linear maps

Section 1.4: Matrix multiplication and map composition

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Matrix product formulas

Definition: Matrix-matrix product

Let A be an $m \times n$ matrix, let B be an $n \times k$ matrix, and write

$$B = [\mathbf{b}_1 \quad \mathbf{b}_2 \quad \cdots \quad \mathbf{b}_k]$$

where $\mathbf{b}_j$ is column j of B . The product AB is the $m \times k$ matrix

$$AB = [A\mathbf{b}_1 \quad A\mathbf{b}_2 \quad \cdots \quad A\mathbf{b}_k].$$

Activity: Computing a matrix product using columns

Let

$$A = \begin{bmatrix} 1 & -3 & 5 \\ 3 & 1 & 4 \end{bmatrix}, \quad B = \begin{bmatrix} 5 & 1 \\ -2 & -6 \\ -1 & 0 \end{bmatrix}.$$

Compute AB .

Theorem

If A is $m \times n$ and B is $n \times k$, then the (i, j) -entry of AB is the dot product of row i of A with column j of B :

$$(AB)_{ij} = \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.$$

Why is this true?

By the definition of matrix multiplication, column j of AB is $A\mathbf{b}_j$, where $\mathbf{b}_j = (b_{1j}, \dots, b_{nj})$ is column j of B . The row dot-product rule for matrix-vector multiplication in Row dot-product view therefore gives

$$\begin{aligned}(AB)_{ij} &= (A\mathbf{b}_j)_i \\ &= \text{row}_i(A) \cdot \mathbf{b}_j \\ &= \sum_{\ell=1}^n a_{i\ell} b_{\ell j}.\end{aligned}$$

Note: Row-column interpretation

The sum above is the dot product of row i of A and column j of B . Viewing both as n -vectors,

$$(AB)_{ij} = (a_{i1}, \dots, a_{in}) \cdot (b_{1j}, \dots, b_{nj}).$$

Pair entries in corresponding positions, multiply, and add.

Activity: Computing a matrix entry

If

$$A = \begin{bmatrix} -3 & -4 & 1 \\ 2 & 4 & 0 \\ 1 & -4 & -5 \end{bmatrix}, \quad B = \begin{bmatrix} -1 & 2 & -4 \\ -4 & -3 & -1 \\ 4 & 3 & 1 \end{bmatrix},$$

what is the $(2, 3)$ -entry of AB ?

Theorem: Properties of matrix multiplication

Assume that A, B, C are matrices of appropriate sizes and r is a scalar. Then

1. $A(BC) = (AB)C$.
2. $(A + B)C = AC + BC$.
3. $C(A + B) = CA + CB$.
4. $A(rB) = r(AB) = (rA)B$.
5. $(AB)^T = B^T A^T$.

Why is this true?

We prove property 5. Let A be $m \times n$ and B be $n \times k$. Both $(AB)^T$ and $B^T A^T$ have shape $k \times m$. Transposition exchanges row and column indices, so the entry formula in the theorem gives

$$\begin{aligned}
((AB)^T)_{ij} &= (AB)_{ji} \\
&= \sum_{\ell=1}^n a_{j\ell} b_{\ell i} \\
&= \sum_{\ell=1}^n b_{\ell i} a_{j\ell} \\
&= \sum_{\ell=1}^n (B^T)_{i\ell} (A^T)_{\ell j} \\
&= (B^T A^T)_{ij}.
\end{aligned}$$

We can reverse the two scalar factors in each summand because real-number multiplication is commutative. Every corresponding entry agrees, so $(AB)^T = B^T A^T$.

Note: Shape habit for matrix products

Before forming AB , check the inner dimensions. If A is $m \times n$ and B is $n \times k$, then the shared inner dimension n makes the product possible, and the outer dimensions give AB shape $m \times k$.

Matrix multiplication as composition

If $T_A(\mathbf{x}) = A\mathbf{x}$ and $T_B(\mathbf{x}) = B\mathbf{x}$, then applying B first and then A gives

$$T_A(T_B(\mathbf{x})) = A(B\mathbf{x}) = (AB)\mathbf{x}.$$

Matrix multiplication represents composition of matrix maps.

Activity: Composition shape check

Suppose A is 4×3 and B is 3×2 . What is the shape of AB , and what is the domain and codomain of the composed map $\mathbf{x} \mapsto A(B\mathbf{x})$?

Activity: Applying one matrix to many points

Put the vertices of a shape in the columns of a matrix X . Then $Y = AX$ contains the transformed vertices.

```
import numpy as np

np.set_printoptions(precision=3, suppress=True)

# Columns are vertices of the unit square.
# The final column repeats the first vertex to close the shape.
X = np.array([
    [0., 1., 1., 0., 0.],
    [0., 0., 1., 1., 0.],
])

A = np.array([
    [1., 1.],
    [0., 1.],
])

Y = A @ X
Y
```

Output:

```
array([[0., 1., 2., 1., 0.],
       [0., 0., 1., 1., 0.]])
```

1. What are the columns of Y ?

Activity: Applying one matrix to many points (continued)

2. Which transformation from the gallery is this?
3. Why does the bottom edge stay fixed?

Warning

In this geometric example, the columns of X are points, so $Y = AX$ applies the same matrix to every point. In many data tables, observations are stored as rows. For example, a document matrix may have one document per row. Always check what rows and columns represent before interpreting a product.

Warning: Later square-grid visualizations

The matrix computation $Y = AX$ applies one matrix to every column of X . For a 2×2 matrix, the images of the coordinate directions determine the whole grid. Later, in Corners are not enough, the same columns-of-points convention is used for rules that are not matrix maps. Then four corners may not tell the whole story. A grid of input points gives more information.

A missing rule: AB need not equal BA

Multiplication of real numbers is commutative: $ab = ba$. Matrix multiplication is not commutative in general. For two square matrices A and B of the same size, we say that A and B commute when

$$AB = BA.$$

Some pairs commute and some do not.

Activity: Commute or not?

For each pair below:

1. compute both products;
2. determine whether $AB = BA$;

Pair A: two distinct diagonal transformations. Use the horizontal stretch and reflection across the y -axis:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad F = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}.$$

Pair B: a shared axis label is not enough. Use the same horizontal stretch and the horizontal shear:

$$S = \begin{bmatrix} 2 & 0 \\ 0 & 1 \end{bmatrix}, \quad H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Pair C: reflection across the line $y = x$. Use the same horizontal stretch and

$$M = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}.$$

Tags. [U1-L06 | P+R | Core]

Activity: Two horizontal shears

Let

$$H = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad G = \begin{bmatrix} 1 & 2 \\ 0 & 1 \end{bmatrix}.$$

The matrix G is a stronger horizontal shear.

1. Justify the claim that applying H twice has the same effect as applying G once.
2. Compute HG and GH . Do the matrices commute?

Tags. [U1–L06]

Note

Some structural patterns settle the question quickly. Two diagonal matrices of the same size commute. For example, if

$$A = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix}, \quad B = \begin{bmatrix} c & 0 \\ 0 & d \end{bmatrix},$$

then

$$\begin{aligned} AB &= \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix}, \\ BA &= \begin{bmatrix} ca & 0 \\ 0 & db \end{bmatrix} = \begin{bmatrix} ac & 0 \\ 0 & bd \end{bmatrix} = AB. \end{aligned}$$

The equality follows because multiplication of real numbers is commutative.

Every square matrix commutes with itself. The two horizontal shears in Two horizontal shears are closely related to this fact: as the solution shows, $G = H^2$, so both HG and GH apply the same matrix H three times. Associativity gives

$$HG = H(HH) = (HH)H = GH.$$

In general, compute AB and BA and compare.

Many token queries and keys

Example: Many token queries at once

Suppose a sequence has L tokens. Store the token vectors as rows of a matrix

$$X = \begin{bmatrix} \mathbf{x}_1^T \\ \mathbf{x}_2^T \\ \vdots \\ \mathbf{x}_L^T \end{bmatrix} \in \mathbb{R}^{L \times d}.$$

A transformer layer forms three views of the same token vectors:

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V.$$

The matrices W_Q, W_K, W_V are weight matrices. Their entries are adjusted during training. For now, treat them as ordinary matrices that assign each token three roles.

For destination position i , the vector $\mathbf{q}_i$ is its query: what that position uses to decide which source positions are relevant to its attention output.

For source position j , the vector $\mathbf{k}_j$ is its key: what is compared with the query from a destination position.

For source position j , the vector $\mathbf{v}_j$ is its value: the information that position can contribute after weights are chosen.

Write

$$Q = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix}, \quad K = \begin{bmatrix} \mathbf{k}_1^T \\ \mathbf{k}_2^T \\ \vdots \\ \mathbf{k}_L^T \end{bmatrix}.$$

Then

$$K^T = [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L].$$

So

$$QK^T = \begin{bmatrix} \mathbf{q}_1^T \\ \mathbf{q}_2^T \\ \vdots \\ \mathbf{q}_L^T \end{bmatrix} [\mathbf{k}_1 \ \mathbf{k}_2 \ \cdots \ \mathbf{k}_L] = \begin{bmatrix} \mathbf{q}_1 \cdot \mathbf{k}_1 & \mathbf{q}_1 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_1 \cdot \mathbf{k}_L \\ \mathbf{q}_2 \cdot \mathbf{k}_1 & \mathbf{q}_2 \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_2 \cdot \mathbf{k}_L \\ \vdots & \vdots & \ddots & \vdots \\ \mathbf{q}_L \cdot \mathbf{k}_1 & \mathbf{q}_L \cdot \mathbf{k}_2 & \cdots & \mathbf{q}_L \cdot \mathbf{k}_L \end{bmatrix}.$$

Thus the score matrix

$$S = QK^T$$

has entries

$$S_{ij} = \mathbf{q}_i \cdot \mathbf{k}_j.$$

Row i of S contains the scores used when computing the attention output at position i .

Activity: Many-token shape and entry check

Suppose

$$X \in \mathbb{R}^{5 \times 4}$$

and

$$W_Q, W_K, W_V \in \mathbb{R}^{4 \times 3}.$$

Let

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of K^T ?
3. What is the shape of $S = QK^T$?
4. Suppose

$$\mathbf{q}_2 = \begin{bmatrix} q_{21} \\ q_{22} \\ q_{23} \end{bmatrix}, \quad \mathbf{k}_4 = \begin{bmatrix} k_{41} \\ k_{42} \\ k_{43} \end{bmatrix}.$$

Expand the entry S_{24} .

5. What does S_{24} compare?
6. What does row 2 of S contain?

For this section, an

attention calculation

has three steps:

dot-product scores $\longrightarrow$ weights $\longrightarrow$ weighted averages.

The score S_{ij} compares the query at destination position i with the key at source position j . A larger score means source position j is more relevant to the attention output at destination position i .

The scores are not yet the output. For each token i , a weighting rule converts row i of S into numbers

$$\alpha_{i1}, \alpha_{i2}, \dots, \alpha_{iL}$$

that are nonnegative and add to 1. These are the

attention weights

for token i . They say how much token i

uses each value vector.

Let

$$A_{\text{att}} = [\alpha_{ij}]$$

be the attention-weight matrix. If the value vectors are stored as rows,

$$V = \begin{bmatrix} \mathbf{v}_1^T \\ \mathbf{v}_2^T \\ \vdots \\ \mathbf{v}_L^T \end{bmatrix},$$

then the attention output is

$$H = A_{\text{att}} V.$$

Row i of H is

$$\mathbf{h}_i^T = \alpha_{i1} \mathbf{v}_1^T + \alpha_{i2} \mathbf{v}_2^T + \dots + \alpha_{iL} \mathbf{v}_L^T.$$

Equivalently,

$$\mathbf{h}_i = \alpha_{i1}\mathbf{v}_1 + \alpha_{i2}\mathbf{v}_2 + \cdots + \alpha_{iL}\mathbf{v}_L.$$

Thus H is the attention-output matrix, and each row of H is a weighted average of value vectors. These rows contribute to later token representations; a full transformer layer combines them with the current representations and processes them further.

Activity: Many-token attention output

Suppose

$$X \in \mathbb{R}^{4 \times 3}, \quad W_Q, W_K, W_V \in \mathbb{R}^{3 \times 2}.$$

Define

$$Q = XW_Q, \quad K = XW_K, \quad V = XW_V, \quad S = QK^T.$$

A weighting step converts S into an attention-weight matrix

$$A_{\text{att}} \in \mathbb{R}^{4 \times 4}.$$

The attention output is

$$H = A_{\text{att}}V.$$

1. What are the shapes of Q , K , and V ?
2. What is the shape of $S = QK^T$?
3. What does S_{ij} compare?
4. What does row i of A_{att} tell you?
5. What is the shape of H ?
6. Why is each row of H a weighted average of value vectors?

Activity: Many-token attention output (continued)

7. Now suppose row 2 of A_{att} is

$$\alpha_2^T = [0 \quad 1/2 \quad 1/4 \quad 1/4]$$

and

$$V = \begin{bmatrix} 1 & 0 \\ 2 & 0 \\ 0 & 4 \\ 4 & 4 \end{bmatrix}.$$

Compute the attention-output vector h_2 .

Warning

In real transformer attention, the weighting rule usually uses scaled and masked scores before forming weights. For Unit 1, the important facts are: rows of A_{att} are weights, the weights add to 1, and $A_{\text{att}}V$ forms weighted averages of the rows of V .

Activity: The same shape check in code

The following code performs a many-token shape check similar to the previous activity.

```
import numpy as np
```

Activity: The same shape check in code (continued)

```
X_tokens = np.ones((5, 4))
WQ = np.ones((4, 3))
WK = np.ones((4, 3))
WV = np.ones((4, 3))

Q = X_tokens @ WQ
K_tokens = X_tokens @ WK
V_tokens = X_tokens @ WV
scores_many = Q @ K_tokens.T

Q.shape, K_tokens.shape, V_tokens.shape, scores_many.shape
```

Output:

```
((5, 3), (5, 3), (5, 3), (5, 5))
```

1. Which line creates all query-key scores at once?
2. Why is `scores_many.shape` equal to `(5, 5)`?